Three distinct geminal couplings

ASN matrix for DSS and homotaurine | Derived and checked 18 October 2026

Result. Use the 4 x 4 parity block below, with three independent geminal couplings and two independent nearest-neighbor coupling differences. It retains type-I transfer and type-II mixing. The 3 x 3 one-triplet truncation does not retain the latter.

Scope. The protons within each CH_2 pair are chemically equivalent; intrapair scalar couplings are retained in full and interpair couplings are secular (I_zI_z). This is the high-field reduced Hamiltonian, without RF during evolution. It is exact within these assumptions, rather than the full isotropic six-proton molecular Hamiltonian.

$$J_A = J_{AA'}, \quad J_M = J_{MM'}, \quad J_X = J_{XX'}, \quad S_J = J_A + J_M + J_X$$

$$d = \Delta J_{AM} = J_{AM} - J_{AM'}, \qquad e = \Delta J_{MX} = J_{MX} - J_{MX'}$$

As in the published model, assume $J_{AM} = J_{A'M'}$ and $J_{AM'} = J_{A'M}$, and analogous equalities for the other pairs. **Here d and e are full differences; each off-diagonal element is d/2 or e/2.** The emailed rotated fit tables instead list the half-differences.

$$|S\rangle = (|\alpha\beta\rangle - |\beta\alpha\rangle)/\sqrt{2}, \quad |T\rangle = |T_0\rangle = (|\alpha\beta\rangle + |\beta\alpha\rangle)/\sqrt{2}$$

Even singlet parity: main ASN block

$$\mathcal{B}_+ = (|TSS\rangle, |STS\rangle, |SST\rangle, |TTT\rangle), \quad C_+ = -3S_J/4$$

$$\mathcal{H}_+ = C_+ I_4 + \begin{bmatrix} J_A & d/2 & 0 & e/2 \\ d/2 & J_M & e/2 & 0 \\ 0 & e/2 & J_X & d/2 \\ e/2 & 0 & d/2 & S_J \end{bmatrix}$$

The fourth row and column contain the type-II couplings. All entries and eigenvalues are in Hz: U(t) = exp[-i 2 pi H t] in this frequency convention. Do not add another 2 pi before numerical diagonalization if you want frequencies in Hz.

Odd singlet parity: complementary block

$$\mathcal{B}_- = (|STT\rangle, |TST\rangle, |TTS\rangle, |SSS\rangle), \quad C_- = S_J/4$$

$$\mathcal{H}_- = C_- I_4 + \begin{bmatrix} -J_A & d/2 & 0 & e/2 \\ d/2 & -J_M & e/2 & 0 \\ 0 & e/2 & -J_X & d/2 \\ e/2 & 0 & d/2 & -S_J \end{bmatrix}$$

For zero non-nearest-neighbor difference, H_- = -D H_+ D - (S_J/2)I, with D = diag(1,-1,1,-1). Thus the two blocks have reflected energy spectra and identical sets of internal energy gaps, even for unequal geminal couplings.

Couplings and parameter ranges

Training bounds are a modeling choice; fitted examples are historical reference points.

Parameter	Recommended initial training range / Hz	Interpretation
J_A, J_M, J_X	[-15.0, -12.5], independently	Agrees with the proposed interval, written in increasing order.
d, e	[0.1, 7.0], independently	Full coupling differences; contains the examples below.
Boundary validation	Also test d = 0 and/or e = 0	Checks the decoupled-chain limits without changing the main sampling range.

These are sensible initial boxes for DSS and homotaurine, not universal chemical limits or confidence intervals. In the nearest-neighbor open chain, signs of d and e can be changed by basis phases without changing the eigenvalues, so positive ranges suffice for spectral training. Include consistent signs when a third, loop-closing coupling difference is retained.

Values recovered from the emailed fit tables

Parameter / Hz	DSS	Homotaurine
J_A	-13.758727	-12.743923
J_M	-13.408159	-13.969277
J_X	-14.121958	-14.109177
d = full Delta J_{AM}	5.269170	3.093406
e = full Delta J_{MX}	6.597217	2.344390
Optional q = Delta J_{AX}	0.002490	0.027306

DSS: 31 July 2026 email, *parameters_298.txt*, labeled 298 K. Homotaurine: internally consistent physical row from the 15 August 2026 email, *Homotaurine_parameters_298.txt*. **The homotaurine filename says 298 K, while its body says 289 K.** The 16 August OST revision changes rotated geminal entries without updating the physical row. Therefore the homotaurine numbers above are illustrative; reconcile the temperature and revised parameter set before using them as a definitive experimental benchmark.

Small non-nearest-neighbor correction

For q = Delta J_{AX}, replace the zeros at (1,3), (3,1), (2,4), (4,2) in **both** blocks by q/2 (indices here start at 1). The Python function supports this optional sixth parameter. It breaks the energy-reflection relation between the parity blocks. For millihertz comparisons, validate both blocks and check the secular approximation against the full spin model.

Molecule / even block	Small gap / Hz	Middle gap / Hz	Sum gap / Hz
DSS, q = 0	3.708572	4.666649	8.375221
DSS, with fitted q	3.707014	4.668631	8.375644
Homotaurine, q = 0	1.967923	2.185440	4.153363
Homotaurine, with fitted q	1.951897	2.203535	4.155432

These are numerical gaps among the upper three eigenvalues of H_+, not a fitted spectrum or a guarantee of observable intensity. For homotaurine, including q changes the middle gap by 0.018095 Hz. With q nonzero the odd block has different gaps; the supplied script also prints them.

ASN inputs and verification

Five physical parameters reduce to four dimensionless inputs when q = 0.

Sample the five physical parameters within the boxes above first, then normalize. This retains physical sampling correlations. Use a positive scale and a documented energy reference:

$$s = -J_M > 0, \qquad E_{\mathrm{ref}} = (-3J_A + J_M - 3J_X)/4$$

$$x_A = (J_A - J_M)/s, \quad x_X = (J_X - J_M)/s, \quad y = d/s, \quad z = e/s$$

$$K = (\mathcal{H}_+ - E_{\mathrm{ref}} I_4)/s$$

$$K = \begin{bmatrix} x_A & y/2 & 0 & z/2 \\ y/2 & 0 & z/2 & 0 \\ 0 & z/2 & x_X & y/2 \\ z/2 & 0 & y/2 & -2 + x_A + x_X \end{bmatrix}$$

Recover eigenvalues as $E_i = E_{\mathrm{ref}} + s\,\mathrm{lambda}_i(K)$, and frequencies as $s\,|\mathrm{lambda}_i - \mathrm{lambda}_j|$. If q varies as a sixth physical parameter, add q/s as a fifth dimensionless input. The supplied four-feature dataset generator assumes q = 0.

Use the supplied Python file

```
python asn_three_ch2.py --check
python asn_three_ch2.py --samples 10000 --csv training_data.csv
```

Only NumPy is required. `hamiltonian(...)` returns the even block by default; set `parity="odd"` for the complementary block and dJAX=q for the optional correction. `make_dataset(...)` returns physical inputs, four dimensionless features, all four sorted dimensionless eigenvalues, three principal gaps, scale, and reference.

Branch handling. Keep all four eigenvalues. A positive-eigenvalue filter is not invariant under an energy shift. In the stated parameter domain, the principal one-triplet frequencies use the upper three levels of the even block; the analogous odd-block frequencies use the lower three. Intensity calculations still require eigenvectors and preparation/readout operators. A polynomial ASN should be assessed near small gaps and avoided crossings, where a low-order global approximation can be difficult.

Audit and independent checks

The uploaded *data_generator.py* already contains the correct generalized even block, ordered as (SST, STS, TSS, TTT); interchange its first and third rows/columns to obtain the ordering on page 1. The idealized driver ties the three geminal parameters together. Its *ham_reduced_antisymmetric* function is not symmetric as written. The separate *data_generator_simplified.py* also fails its own quoted idealized-frequency check. Use the checked module supplied here.

Passed checks: all six operator coefficients in both blocks were projected from independent 64-dimensional six-proton spin operators using exact integer arithmetic; invariance of each subspace was also proved by the coefficient checks. Numerical checks covered 100 random parity/normalization/scaling cases, zero-difference limits, the equal-coupling reference, reconstruction of dataset targets, and agreement with the original correct even block after basis reordering. A 10,000-sample generation smoke check also passed.

Sources: K. F. Sheberstov, *Aliphatic chains as physical realizations of one-dimensional spin chains supporting magnetically silent spin waves*, Science Advances 12, eaef7347 (2026), DOI: 10.1126/sciadv.aef7347; supplied Python files; DSS fit email (31 July); homotaurine fit email (15 August); OST revision (16 August). The PRX Intelligence editorial email (14 October) requests real-molecule NMR predictions, stronger literature context, and comparisons to state-of-the-art models.